Implemented Prototype Architecture of the Land Management Blockchain System (LMBS)

Four layers with cross-cutting security controls. **Solid boxes are implemented** in the prototype; the dashed box on the right lists planned work. Numbered badges ①–⑦ trace one registration end to end.

Users / Institutional Actors

Applicant / Citizen · Title Holder · Registrar · Institutional Approver · Identity Issuer · Dispute Officer · Encumbrance Officer · Administrator

Layer 1 – User and Institutional Application Layer

React.js web portal · English / Sesotho

Registration / Transfer Wizard
- submit registration
- submit transfer
- upload documents (encrypted)
- save / resume drafts

Institutional Approvals
- approve / reject request
- signed vote, offline (EIP-712) NEW
- execute request

Blockchain Registry Console
- titles and requests
- disputes / encumbrances
- identity status
- cancel own request

Title Verification & Search NEW
- certificate checked on-chain
- parcel search
- provenance timeline
- network status

Wallet Panel
- wallet linking (signature only)
- **user signs every on-chain tx**
- on-chain role display
- no seed phrase or private key ever requested

① ④ REST / JSON + JWT — user-signed tx: submit / approve / execute

Layer 2 – API, Security and Workflow Services Layer

Node.js / Express backend · REST + JWT

Document Security
keccak256 plaintext hash · AES-256-GCM encryption · key from server environment (`ENCRYPTION_KEY`)

Authentication & RBAC
JWT authentication · bcrypt password hashing · application roles enforced per route

Wallet Verification
challenge → signature → nonce check → wallet bound to user account

Workflow / API Service
request workflow, validation and business rules; off-chain records never change ownership

Blockchain Verification
fetches tx receipt and checks the `RequestSubmitted` event; client-reported hashes are never trusted

Certificate & Provenance NEW ⑦
certificate valid only if `ownerOf()` and document hash match the chain · history built from read models

Audit, Notifications & Network NEW
audit log of every state-changing request · notifications (email/SMS stubbed) · Besu peer & block status

Blockchain Indexer
polls LandRegistry, Dispute and Encumbrance events · idempotent · resumes from `chain_cursor`

③ read receipt + events (JSON-RPC) — on-chain events ⑥ — ② user-signed tx: submit / approve / execute

Layer 3 – Permissioned Besu Blockchain and Smart Contract Layer

Hyperledger Besu QBFT Network

Validator 1	Validator 2	Validator 3	Validator 4
Land Administration Authority	Dept. of Survey & Physical Planning	Deeds Registry	Ministry of Local Government

Chain ID: 2026 · Block period: 2 s · Validators: 4
QBFT quorum: 3 · Fault tolerance: 1 · Zero gas price
Node + account allowlists · Besu 25.9.0 in Docker Compose

Smart Contracts · Solidity 0.8.24 · OpenZeppelin · Hardhat

Access & Governance
- `RoleAuthority.sol` — OpenZeppelin AccessControl
- `Roles.sol` — 6 roles
- `AuthorityGated.sol` NEW — base contract: every check → `hasRole`

Identity
- `IdentityRegistry.sol` — register · status · isVerified
- `credentialHash`, `issuerRef`: hooks for future VCs

Land Constraints
- `DisputeRegistry.sol` — open / close dispute
- `EncumbranceRegistry.sol` — register / discharge

Core Land Registry ⑤
- `LandRegistry.sol` — ERC-721 + EIP-712
- registration and transfer requests
- parcel reservation (no duplicate parcels)
- M-of-N approvals: transaction *or* signed vote (nonce, deadline, chain id)
- `execute()` checks `executionBlocker()`
- direct token transfer disabled: a title moves only via `execute()` NEW

checked by executionBlocker() · hasRole()

⚖ Two distinct mechanisms

Business approval: M-of-N institutional approval in LandRegistry (default 3 approvals; 2 rejections close a request). Threshold snapshotted at submission.

Blockchain consensus: QBFT agreement among Besu validators (quorum 3 of 4).

⟳ Request lifecycle

```
PENDING → REJECTED
PENDING → CANCELLED
PENDING → APPROVED → EXECUTED
```

An approved request stays APPROVED while a blocker holds:
`OPEN_DISPUTE` `BLOCKING_ENCUMBRANCE` `PARTY_NOT_VERIFIED` `OWNER_CHANGED`

ciphertext + hash — Indexer writes read models

Layer 4 – Off-Chain Evidence and Read-Model Layer

Encrypted file store · MongoDB

📄 Encrypted Document Store
- supporting documents, AES-256-GCM on disk
- plaintext keccak256 anchored on-chain
- verify endpoint re-checks the hash against the chain
- download only for uploader or officer roles

🗄 MongoDB Read Models (written only by the Indexer)

`chain_requests` `parcels` `disputes` `chain_cursor` `title_deeds` `transfers` `encumbrances`

The chain is the authority; these collections serve queries and dashboards.

🗄 MongoDB Application Data

`users` `audit_log` `documents` `notifications`

accounts (bcrypt), document metadata, audit trail

🛡 Cross-cutting Security & Controls

IDENTITY & ACCESS
- JWT + RBAC on every API route
- bcrypt password hashing
- wallet challenge–signature binding
- on-chain RBAC via RoleAuthority
- separation of duties between institutions

TRANSACTION INTEGRITY
- EIP-712 votes: nonce, deadline, chain id
- tx receipts and events verified server-side
- signing keys stay in the user's wallet
- Besu node + account allowlists

DATA PROTECTION
- AES-256-GCM documents; only the hash goes on-chain
- helmet headers, CORS allowlist; rate limiting on login and sign-up
- audit log of state-changing requests (secrets redacted)

Registration flow

1. Applicant uploads documents; backend encrypts them and returns the keccak256 hash
2. Applicant signs `submitRegistration` in their own wallet
3. Backend verifies the receipt and `RequestSubmitted` event
4. Institutions approve (transaction or signed EIP-712 vote)
5. `execute()` checks identity, disputes and encumbrances, then mints the ERC-721 title
6. Indexer projects the events into MongoDB read models
7. Certificate is verified against `ownerOf()` on-chain

Planned / out of scope

- W3C verifiable credentials and DIDs
- BBS+ selective disclosure, zero-knowledge proofs
- KMS / HSM key management and key rotation
- MongoDB client-side field encryption
- live email / SMS gateways
- GIS and cadastral system integration

Legal note: the ERC-721 title token is an internal digital record; legal ownership remains governed by national land law and the authorised land registration process. · **Tooling:** Hardhat contract tests, Jest backend tests, full-stack E2E script.

call / data flow · user-signed transaction · on-chain events · constraint check